

MINFENG, China

WMO: 518390

Lat: 37.067N

Lon: 82.717E

Elev: 4624

StdP: 12.40

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.9	(c) 10.0	(d) -7.5	(e) 4.4	(f) 25.8	(g) -5.1	(h) 5.0	(i) 24.3	(j) 12.1	(k) 28.3	(l) 10.0	(m) 29.7	(n) 0.6	(o) N/A	(p) 0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.0	(c) 96.0	(d) 63.6	(e) 93.3	(f) 62.7	(g) 90.7	(h) 62.0	(i) 67.5	(j) 88.4	(k) 65.7	(l) 87.0	(m) 64.1	(n) 84.8	(o) 4.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 60.7	(b) 94.4	(c) 75.6	(d) 58.7	(e) 87.8	(f) 74.2	(g) 56.8	(h) 81.6	(i) 73.4	(j) 34.9	(k) 88.6	(l) 33.2	(m) 87.5	(n) 31.9	(o) 84.6	(p) 75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.0	9.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			0.7	72.1	3.3	2.1	-1.6	73.6	-3.5	74.8	-5.4	76.0	-7.8	77.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	23.4	31.5	45.0	58.4	67.6	74.0	76.5	75.5	65.9	52.7	39.4	26.6		
	DBStd	19.60	5.79	5.63	7.19	6.17	5.93	5.09	4.54	4.62	4.81	5.98	6.87	6.50		
	HDD50	2626	825	518	186	7	0	0	0	0	0	41	323	726		
	HDD65	5493	1290	938	620	211	41	3	0	1	46	383	769	1191		
	CDD50	3773	0	0	31	260	545	721	823	790	476	124	3	0		
	CDD65	1165	0	0	0	14	121	274	358	326	71	1	0	0		
	CDH74	14520	0	0	1	395	1582	3163	4146	4000	1129	104	0	0		
	CDH80	6199	0	0	0	104	525	1354	1965	1935	307	9	0	0		
(14) Wind	WSAvg	3.3	2.5	3.2	3.7	4.4	4.2	4.1	3.7	3.2	3.1	2.8	2.5	2.2		
(15) Precipitation	PrecAvg	2.0	0.1	0.0	0.1	0.1	0.2	0.6	0.4	0.2	0.2	0.0	0.0	0.0		
	PrecMax	5.6	0.3	0.3	0.4	1.1	1.0	2.6	1.4	2.2	1.7	0.1	0.1	0.2		
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.1	0.1	0.1	0.1	0.2	0.2	0.6	0.4	0.4	0.4	0.0	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	57.3	71.6	88.0	92.1	99.1	99.4	98.9	91.1	81.2	67.5	52.9		
		MCWB	32.7	38.1	45.7	54.1	58.0	62.2	64.0	65.5	62.0	53.1	44.8	36.4		
	2%	DB	41.3	51.1	68.1	82.6	89.1	93.8	96.3	95.3	86.4	77.2	61.4	45.3		
		MCWB	29.7	35.8	44.3	52.4	57.4	62.0	63.4	64.3	58.8	51.3	41.9	32.9		
	5%	DB	37.7	47.5	64.9	78.5	85.8	90.5	93.4	92.8	83.6	72.6	57.2	40.9		
		MCWB	27.9	33.4	43.1	50.4	56.1	61.5	63.5	62.9	57.4	49.3	39.9	30.3		
	10%	DB	34.6	44.4	60.9	74.4	83.1	87.3	90.2	89.9	80.9	68.3	52.9	37.4		
		MCWB	26.0	31.6	41.1	48.4	55.2	59.9	62.8	62.5	56.0	47.4	37.8	28.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	39.1	46.7	55.1	61.2	67.7	70.1	70.9	65.3	54.9	45.8	36.8		
		MCDB	46.0	55.8	69.3	86.7	84.9	90.1	88.2	90.9	85.1	75.3	64.2	51.4		
	2%	WB	30.0	36.0	44.8	53.0	59.0	64.7	67.3	68.0	60.8	52.4	42.6	33.1		
		MCDB	40.3	49.9	67.0	80.5	83.5	87.1	87.7	89.0	81.7	74.6	59.5	44.7		
	5%	WB	27.9	34.0	43.2	51.0	57.6	62.8	65.7	65.7	59.1	50.1	40.3	30.6		
		MCDB	37.1	46.2	64.0	76.5	82.3	84.9	86.5	87.3	79.9	71.7	55.9	40.0		
	10%	WB	26.1	31.9	41.4	48.9	56.1	61.2	64.4	63.8	57.4	48.0	38.3	28.5		
		MCDB	34.2	43.5	59.8	72.5	79.8	83.4	84.9	84.8	77.4	66.9	52.4	37.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	21.8	24.5	25.7	23.9	23.0	23.0	26.0	28.4	30.0	25.3	20.6		
		MCDBR	25.8	27.0	30.3	35.3	31.5	30.3	31.2	33.4	32.9	36.2	31.3	26.0		
		MCWBR	17.1	15.8	14.6	14.2	12.1	11.7	10.9	11.8	12.2	15.3	16.6	16.2		
	5% WB	MCDBR	24.9	25.3	29.3	31.3	25.7	24.2	22.9	26.9	28.3	33.6	30.7	24.8		
		MCWBR	16.8	14.9	14.3	12.9	10.7	10.7	9.8	12.3	12.4	14.6	16.4	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.383	0.491	0.590	0.624	0.648	0.639	0.640	0.577	0.430	0.354	0.327		
	taud		2.146	1.969	1.702	1.528	1.515	1.525	1.569	1.556	1.631	1.878	2.065	2.124		
	Ebn at Noon		268	262	242	226	222	216	217	212	215	241	251	255		
	Edh at Noon		36	49	70	89	91	90	86	85	75	53	38	34		
(44) All-Sky Solar Radiation	RadAvg		925	1203	1646	1949	2154	2117	2042	1909	1733	1499	1086	868		
	RadStd		91	68	86	80	108	100	157	142	87	34	53	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page